

Bechir Sayadi

Azure Cloud Engineer | DevOps & Infrastructure | Cloud Security

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Dresden, Germany (open to relocation) | TU Dresden, B.Sc. Informatik, graduating Sep 2026
Website: <https://bach47-lab.github.io/>



Final-year Computer Science student at TU Dresden (B.Sc. Informatik, graduating Sep 2026) specializing in Microsoft Azure and DevOps automation. I've turned coursework in network security, cryptography, and cloud/edge computing into hands-on infrastructure work and solid foundations in OS, deploying Kubernetes workloads, provisioning Azure resources with Terraform and Bicep, and designing secured, segmented networks end to end. Comfortable across the stack, from IAM and TLS to CI/CD pipelines and monitoring, with working knowledge of AWS and GCP alongside my Azure focus.

TECHNICAL SKILLS

Azure	VMs · App Service · Azure Functions · AKS · ACR · Blob Storage · Azure SQL · VNet · NSG · Azure Firewall · Application Gateway · Key Vault · Azure Monitor
IaC & CI/CD	Terraform (AzureRM) · Bicep · ARM Templates · Azure CLI · PowerShell · Azure DevOps Pipelines · GitHub Actions · Docker · Kubernetes · Helm
Security	Entra ID · RBAC · Managed Identities · Defender for Cloud · Key Vault · TLS/PKI · iptables/NAT · Zero Trust concepts
Monitoring	Azure Monitor · Log Analytics · Application Insights · Prometheus · Grafana
Other Cloud	AWS (EC2, S3, IAM, Lambda) and GCP (Compute Engine, GKE) — working knowledge alongside Azure focus
Languages	Python · Java · Bash · PowerShell · C/C++ · YAML · SQL

KEY PROJECTS

Lunar Edge — Cloud-Hosted Digital Twin for IoT Edge Robotics · *TU Dresden Team Project · Team of 6 · 3 sub-teams* 2026

my sub-team: Digital Twin

- ▶ Deployed Eclipse Ditto via Docker Compose on a university research-cloud VM as the digital twin platform, configuring policy management and Web of Things (WoT) Thing Descriptions
- ▶ Built a Flask REST gateway ingesting live robot telemetry into Ditto, persisted via MongoDB, running as systemd services for 24/7 uptime independent of any developer's machine
- ▶ Implemented a SciPy-based obstacle-detection pipeline over SLAM map data and a live HTML5/JS dashboard visualizing robot position and the map in real time

Enterprise Network Security Design · *TU Dresden Team Project · Network Security Lab · Team of 5* 2026

(lab passed at the university)

- ▶ Designed and deployed a segmented network (router/NAT/firewall, DMZ, internal LAN) across six virtualized Linux hosts (grml)
- ▶ Configured a BIND9 DNS server and an Apache2 web server secured with a Let's Encrypt TLS certificate; validated hardening with nmap port scans
- ▶ Implemented layered iptables NAT/DNAT rules enforcing least-privilege traffic flow, plus automated, encrypted, append-only backups (restic) protecting against ransomware

Flower Shop / Movie Store Full-Stack App (University Module) · *TU Dresden Team Project Team of 7* 2024 - 2025

- ▶ Developed a full-stack web application using Java, Spring Boot, Thymeleaf, and SQL
- ▶ Designed a RESTful API architecture and implemented business logic and database access layers
- ▶ Collaborated in a team using Git branching strategies and agile task management

Microservices Deployment on Azure Kubernetes Service · *Personal* 2024 - 2025

- ▶ Provisioned the full stack in Terraform — AKS cluster, Azure Container Registry, Azure SQL, VNet with proper NSG rules — with zero manual portal steps
- ▶ Built an Azure DevOps pipeline that builds and pushes the Docker image and deploys to AKS via Helm, with Azure Monitor and Application Insights for visibility

Azure Environment with Bicep — Security Fundamentals · *Personal* 2025

- ▶ Deployed a Bicep-based environment — resource groups, VNet, Key Vault for secrets, and Managed Identities instead of hardcoded credentials
- ▶ Enabled Defender for Cloud and worked through its recommendations to raise the environment's Secure Score

TLS Handshake Implementation + Azure API Management · [University-inspired](#)

2024

- Implemented a simplified TLS handshake in Java — RSA key exchange, Diffie-Hellman, AES — from the Cryptography module at TU Dresden
- Deployed the service behind Azure API Management using TLS certificates issued from Key Vault

WORK EXPERIENCE

Student Research Assistant (SHK) — Civil Engineering · [TU Dresden](#)

02.2025 – 07.2025

- Wrote a C++/Qt application reading from multiple measurement sensors in parallel and storing data in structured files for analysis; Git, code reviews, documentation

Student Research Assistant (SHK) — Mechanical Engineering · [TU Dresden](#)

11.2024 – 03.2025

- Built a Python/Qt tool to automate a measurement workflow previously done by hand, working in a mixed team of engineers and CS people

After the SHK positions I chose not to take on additional student jobs — I wanted to focus fully on finishing my degree and spending the remaining time building real Azure, network security, and cloud computing projects, which felt more relevant to where I want to go professionally.

EDUCATION

B.Sc. Informatik (Computer Science)

2023 – present

[Technische Universität Dresden](#)

- Komplexpraktikum Service & Cloud Computing: working on digital twin
- Network Security: in-depth practical lab work on DMZ and Firewalls
- Cryptography: Concepts like Steganographie, Mix- & DC-Nets, Encryption and Digital Signature
- Advanced Computer Networks: TCP/IP deep-dive, routing protocols, SDN and more
- Core modules: Algorithms & Data Structures, Operating Systems, Data Base, Software Engineering

Bachelor Thesis · [Supervisor: Prof. Dr. Ivan Gudymenko, Chair of Privacy and Security in Ubiquitous Computing](#) 2026

"A Comparative Analysis and Lifecycle-Oriented Tool Support for Threat Modeling in Cloud Environments with Privacy and Sovereignty Considerations"

- Benchmarked cloud threat modeling across security, privacy (LINDDUN), and digital sovereignty dimensions against CIS, NIST, ISO, and CSA CCM frameworks; built a proof-of-concept tool covering the full threat lifecycle from identification to mitigation planning

Abitur — Technische Wissenschaften (Technical Sciences)

2022

[Tunesien](#) | *Final Grade: 1.9*

LANGUAGES & Courses (IN PROGRESS)

Arabic C2 · German C1 · English C1 · French C1

AZ-900 Azure Fundamentals · AZ-104 Azure Administrator · AZ-500 Azure Security Engineer · AZ-400 DevOps Engineer (planned exams)

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